

and thus, the two bodies will not move.

A1

- 4 (a) Within the limit of proportionality, the extension/compression of a material is directly proportional to the force applied on it B1
- (b)(i) The centripetal force on the ball in Fig 4.2 is equal the weight of the ball as the tension in the string is zero. B1
- (b)(ii) At the top of the motion, only weight provides for the centripetal force.

$$mg = mr_1\omega^2$$

$$\omega = \sqrt{\frac{g}{r_1}} = \sqrt{\frac{9.81}{0.13}} = 8.687 \text{ rad s}^{-1} \quad \text{B1}$$

$$\text{Spring constant, } k = 5 / (0.148 - 0.13) = 278 \text{ N m}^{-1} \quad \text{B1}$$



[Turn over]

(Since the mass is moving at constant angular velocity, at the bottom, ω is also 8.687 rad s^{-1} .)

Hence, at the bottom, tension – mg provides for centripetal force.

$$T - mg = mr_2\omega^2$$

B1

$$ke - mg = mr_2\omega^2$$

$$(277.8)(L - 0.13) - (5.0) = (5 / 9.81)(L)(8.687^2)$$

C1

$$\text{Solving, } L = 0.172 = 17.2 \text{ cm}$$

A1